

## Contents

### 1 最短路徑

1.1 basic . . . . . 1 43

### 2 圖論

2.1 凸包問題 . . . . . 1 44

### 3 數學

3.1 公式 . . . . . 1 47

## 1 最短路徑

### 1.1 basic

```
1 // c++ code
2 #include <bits/stdc++.h>
3 using namespace std;
4
5 int main() {
6     // test comment
7     cout << "test string\n";
8 }
```

## 2 圖論

### 2.1 凸包問題

```
1 struct node{
2     int x,y;
3     bool operator<(const node &other) const{
4         return (x<other.x) || (x==other.x &&
5             y<other.y);
6     };
7
8     struct ConvexHull{
9         vector<node> vec;
10        int n;
11        void init(vector<node> &v){
12            vec=v;
13            n=v.size();
14        }
15
16        int cross(node a,node o,node b){
17            return
18                (a.x-o.x)*(b.y-o.y)-(b.x-o.x)*(a.y-o.y);
19        }
20
21        vector<int> point;
22        vector<int> area;
23
24        void build(){
25            point=vector<int>(n+1);
26            area=vector<int>(n+1);
27            vector<node> up,down;
28            int upsum=0;
29            int downsum=0;
30            for(int i=0;i<n;i++){
31                while(up.size()>=2 &&
32                    cross(up[up.size()-2],up[up.size()-1],vec[i])<=0){
33                    upsum-=cross(up[up.size()-2],{0,0},up[up.size()-1]);
34                    up.pop_back();
35                }
36                up.push_back(vec[i]);
37                if(up.size()>1)
38                    upsum+=cross(up[up.size()-2],{0,0},up[up.size()-1]);
39
40                while(down.size()>=2 &&
41                    cross(down[down.size()-2],down[down.size()-1],vec[i])>=0){
42                    downsum-=cross(down[down.size()-2],{0,0},down[down.size()-1]);
43                    down.pop_back();
44                }
45                down.push_back(vec[i]);
46                if(down.size()>1)
47                    downsum+=cross(down[down.size()-2],{0,0},down[down.size()-1]);
48
49                area[i+1]=abs(upsum-downsum);
50
51                if(i+1==1) point[i+1]=1;
52                else point[i+1]=up.size()+down.size()-2;
53            }
54
55            double getarea(int p){
56                return area[p]/2.0;
57            }
58
59            int getnum(int p){
60                return point[p];
61            }
62        };
63    };
64 }
```

## 3 數學

### 3.1 公式

• 中文測試

$$\cdot \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$